Climate inequality

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Job market paper

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Abstract

This paper studies the effect of anthropogenic climate change on inequality within countries and globally, an essentially inactive area of economic research despite its intersecting two of the defining challenges of the 21st century. This is accomplished by first extending tools developed in recent macroeconometrics literature to overcome long-standing critical limitations of prevailing methods favored in studies of climate impacts concerning empirical design, efficiency, and sensitivity to misspecification. In an expository application, I demonstrate how several attractive features of these semi-parametric methods enable credible, robust, and directly testable characterization of persistence dynamics which are known to be extremely sensitive to unfalsified parametric assumptions using prevailing methods. I then apply this refined apparatus to rich and newly available distributional national accounts data to document new evidence that temperature shocks exacerbate income inequality within countries, an effect largely driven by disproportionate reduction of absolute and relative incomes of the poorest segments of already-poor countries. To better understand the implications of this incidence pattern for the welfare economics of climate change, I integrate these results with data on observed income distributions and output from the newest

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